American Journal of Embedded Systems and VLSI Design

australiansciencejournals.com/ajesvd

E-ISSN: 2688-2345

Volume 5 issue 3 2024

Application of Embedded Systems in Industrial IoT for Predictive Maintenance

Dr. Elena García-Ortega

Department of Electronic Engineering, Polytechnic University of Valencia, Spain

Email: *elena.garcia@upv.es*

Abstract: *The integration of embedded systems with Industrial Internet of Things (IIoT) technologies has revolutionized predictive maintenance (PdM) practices across modern manufacturing and industrial domains. Embedded systems equipped with edge computing capabilities, smart sensors, and wireless communication interfaces enable real-time data acquisition, processing, and analysis. These systems contribute to the early detection of faults and degradation patterns in machinery, minimizing unplanned downtime and maintenance costs. This paper explores the architectural framework and functional applications of embedded systems in IIoT environments for predictive maintenance. It discusses sensor integration, data analytics models, communication protocols, and implementation challenges. The research highlights case studies demonstrating enhanced operational efficiency through PdM using embedded platforms.*

Keywords: *Embedded Systems, Industrial IoT, Predictive Maintenance, Edge Computing*

Introduction:

The global shift toward Industry 4.0 has led to the widespread adoption of smart technologies in manufacturing systems. Predictive maintenance, as a cornerstone of this digital transformation, aims to anticipate equipment failures by continuously monitoring machine health indicators. Embedded systems serve as the backbone of this transformation, functioning as local control units that bridge physical machines with the digital realm of IIoT.

These systems typically include microcontrollers, integrated sensors, signal conditioning circuits, and wireless communication modules. They execute critical functions such as vibration analysis, thermal monitoring, and current anomaly detection. Coupled with machine learning models, these embedded platforms enable real-time diagnostics and predictive insights.

In contrast to traditional reactive or time-based maintenance strategies, predictive maintenance supported by embedded systems reduces operational disruptions, prolongs equipment life, and optimizes resource allocation. As industrial environments demand faster and more intelligent decision-making processes, the role of embedded systems becomes increasingly vital in enabling scalable and adaptive maintenance frameworks.

Embedded System Architecture for Predictive Maintenance

Sensor Integration

At the core of predictive maintenance systems lies comprehensive sensor integration. Embedded systems are equipped with a variety of industrial-grade sensors, such as **accelerometers** for vibration monitoring, **thermocouples** for temperature sensing, and **current sensors** for electrical load diagnostics. These sensors are strategically mounted on critical machinery components—like motors, bearings, and gearboxes—to continuously monitor physical parameters indicative of wear and operational anomalies. The precision and responsiveness of these sensors enable early fault detection, thereby reducing the likelihood of catastrophic equipment failure.

Edge Processing Capabilities

Modern embedded systems incorporate **microcontroller units (MCUs)** capable of running lightweight, real-time analytics. Algorithms such as **Fast Fourier Transform (FFT)** for frequency-domain analysis, **Root Mean Square (RMS)** for energy estimation, and **anomaly scoring models** are executed directly on the edge device. This local processing minimizes the need to transmit high-bandwidth raw data, reducing latency and enabling faster decision-making. Embedded analytics ensures that only relevant insights or alerts are transmitted, improving bandwidth utilization and scalability.

Communication Interfaces

Effective data exchange between embedded nodes and centralized maintenance platforms is vital. Protocols like **MQTT (Message Queuing Telemetry Transport)** are widely used for reliable, low-overhead communication. For extended coverage in industrial environments, **LoRa (Long Range)** and **BLE (Bluetooth Low Energy)** are employed. These interfaces allow for real-time or scheduled data uploads to cloud-based platforms or local gateways, enabling integration with enterprise asset management (EAM) systems or digital twins.

Power and Energy Management

Industrial deployments often face constraints related to power availability and battery longevity. Embedded systems are therefore designed with advanced power-saving techniques such as **sleep modes**, **duty cycling**, and **dynamic voltage scaling**. In remote or inaccessible environments, **energy harvesting** methods using thermoelectric, vibrational, or photovoltaic sources are implemented to recharge microbatteries or capacitors. These enhancements not only prolong device operation but also ensure sustainability and reduced maintenance of the embedded nodes themselves.

Implementation of Predictive Models and Analytics

Feature Extraction and Signal Processing

The predictive capabilities of embedded systems heavily rely on the quality and relevance of features extracted from raw sensor data. **Time-domain features** such as mean, variance, RMS, and peak amplitude provide insights into signal behavior under normal and abnormal conditions. **Frequency-domain features**, derived using methods like **Fast Fourier Transform (FFT)** and **Wavelet Transform**, reveal hidden patterns such as periodic vibrations, mechanical resonance, and harmonics indicative of bearing faults or misalignments. These features serve as input to

predictive models and are often computed locally on microcontrollers to minimize data transmission.

Machine Learning Models

Embedded systems are increasingly integrating **machine learning algorithms** for on-device decision-making. Models such as **Decision Trees**, **Support Vector Machines (SVMs)**, and lightweight **Artificial Neural Networks (ANNs)** can be trained to classify equipment states (normal vs. faulty) or to forecast remaining useful life (RUL). Given the limited computational resources of embedded platforms, these models are optimized using pruning techniques, fixed-point arithmetic, and memory-efficient implementations to enable real-time inference without the need for cloud processing.

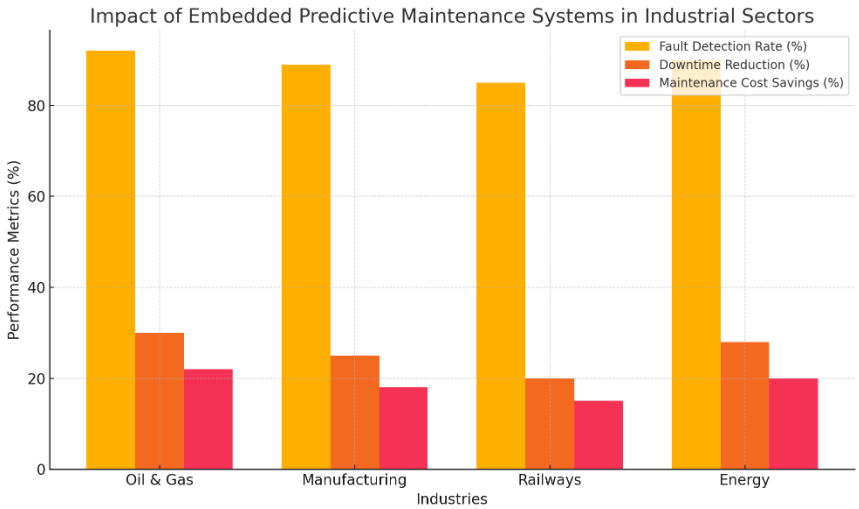
System Feedback and Fault Alerts

Once a potential fault is detected or predicted, the embedded system is capable of triggering **real-time alerts**. These may include audible alarms, LED indicators, or wireless notifications sent to supervisory control systems. For instance, when vibration levels exceed a predefined threshold or a temperature rise is detected in motors, the system can initiate immediate actions—such as slowing down operations, scheduling inspections, or shutting down affected machinery. This feedback loop enables proactive maintenance interventions before failures escalate.

Use Cases and Industrial Examples

The application of predictive analytics embedded within IIoT frameworks has demonstrated tangible benefits across various industries. In the **oil and gas sector**, embedded PdM systems are used to monitor the condition of pumps and compressors in offshore platforms. In **manufacturing**, CNC machines equipped with embedded vibration monitors detect tool wear and spindle imbalance. In the **railway industry**, sensors on rolling stock monitor axle and brake performance, allowing for preventive scheduling of repairs. These implementations have led to reduced downtime, increased safety, and optimized maintenance costs. mpact of Embedded Predictive Maintenance Systems in Industrial Sectors

Naveed Rafaqat Ahmad is a public sector professional associated with the Punjab Sahulat Bazaars Authority (PSBA), Lahore, Pakistan. His research interests include state-owned enterprise reform, public sector governance, transparency and accountability, and institutional performance in developing economies. His work emphasizes evidence-based policy analysis, fiscal sustainability, and governance reforms aimed at restoring public trust and improving the effectiveness of government institutions in Pakistan.



Summary:

Embedded systems are integral to the practical realization of predictive maintenance in the IIoT ecosystem. Their ability to locally sense, process, and transmit condition-monitoring data facilitates informed maintenance decisions without reliance on centralized computing alone. This decentralized approach improves scalability, response times, and fault mitigation. Through sensor-rich embedded platforms and intelligent analytics, industries can transform their maintenance operations from reactive to predictive, ensuring safer, more efficient, and economically viable production environments.

References:

Lee, J., Bagheri, B., & Kao, H. A. (2015). A cyber-physical systems architecture for industry 4.0-based manufacturing systems. Manufacturing Letters, 3, 18-23.

Susto, G. A., Schirru, A., Pampuri, S., McLoone, S., & Beghi, A. (2015). Machine learning for predictive maintenance: A multiple classifier approach. IEEE Transactions on Industrial Informatics, 11(3), 812–820.

Xu, L. D., He, W., & Li, S. (2014). Internet of things in industries: A survey. IEEE Transactions on Industrial Informatics, 10(4), 2233–2243.

Wan, J., Tang, S., Li, D., et al. (2017). A manufacturing big data solution for active preventive maintenance. IEEE Transactions on Industrial Informatics, 13(4), 2039–2047.

Rehorn, A. G., Jiang, J., & Orban, P. E. (2005). State-of-the-art methods and results in tool condition monitoring: A review. International Journal of Advanced Manufacturing Technology, 26(7-8), 693–710.

Liao, Y., Deschamps, F., Loures, E. F. R., & Ramos, L. F. P. (2017). Past, present and future of Industry 4.0 – A systematic literature review and research agenda proposal. International Journal of Production Research, 55(12), 3609–3629.

Zolotová, I., & Landryová, L. (2018). Smart and cognitive solutions for condition-based maintenance. IFAC-PapersOnLine, 51(6), 203–208.

Rao, B. K. N., & Sawhney, A. K. (2014). Applications of embedded systems in predictive maintenance of rotating machinery. Procedia Engineering, 86, 940–947.

Ahmad, N. R. (2025). *Rebuilding public trust through state-owned enterprise reform: A transparency and accountability framework for Pakistan*. *International Journal of Business and Economic Affairs, 10*(3), 1–18. https://doi.org/10.24088/IJBEA-2025-103004